Date: 09 September 2022

BRAC UNIVERSITY

Department of Computer Science Engineering
FINAL EXAMINATION, Summer 2022
CSE 350: Digital Electronics and Pulse Technique

A

Total Marks: 80 (20×4)
Time:2 hours

- *Total no of questions are* ***Six (6).*** *Answer any* ***Four (4)****.*
- *Figure in bracket [ ] next to each question indicates marks for that question.*
- *Write your set number on top of your answer sheet*

Name:	**ID:**	**Section:**

Question No. 1

For the ECL NOR gate in figure 1, ignore the base currents and assume Reference voltage, V_R will be the **average** of logic high and logic low values, $\boldsymbol{V_{BE}(active) = 0.7\ V}$ **and** $\boldsymbol{V_D(diode) = 0.7\ V}$

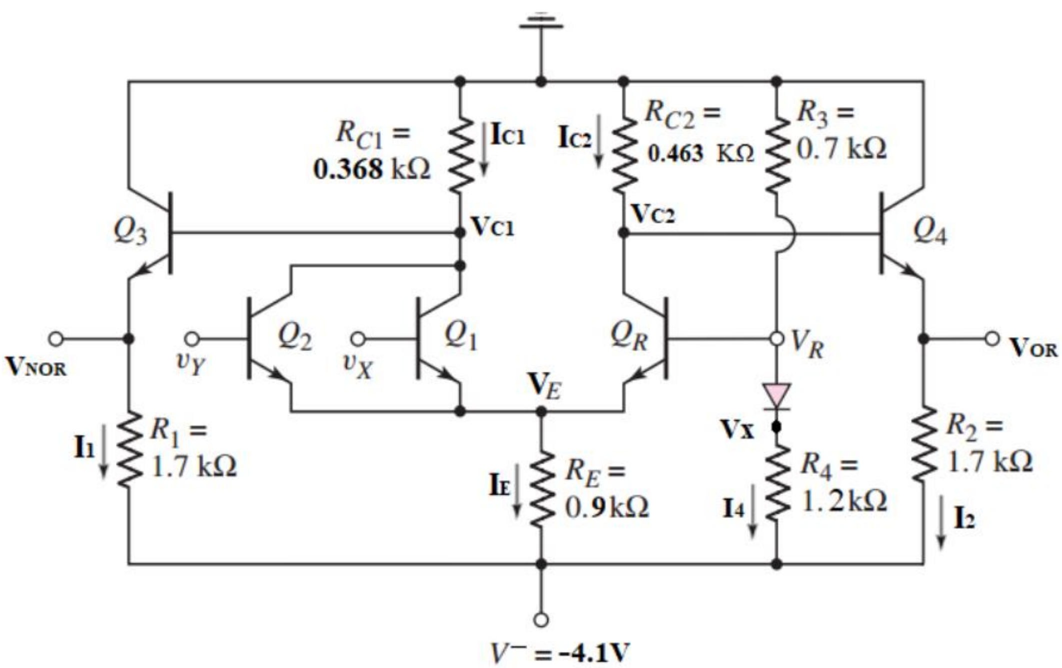

Figure 1

(a)	CO1	Determine the output voltages $\boldsymbol{V_{OR}}$ $\boldsymbol{and}$ $\boldsymbol{V_{NOR}}$ when inputs $\boldsymbol{V_X}$ $\boldsymbol{and}$ $\boldsymbol{V_Y}$ $\boldsymbol{are}$ at Logical High. [Hint: In this condition, $\boldsymbol{V_{OR}}$ is at Logical High voltage and $\boldsymbol{V_{NOR}}$ is at Logical Low voltage]	[10]
(b)	CO2	Utilize the values found in (a) to calculate $\boldsymbol{V_R, V_X, I_1, I_2}$ $\boldsymbol{and}$ $\boldsymbol{I_4}$.	[10]

Question No. 2

The circuit in Figure 2 has the following parameters: $\boldsymbol{\beta_F = 25}$, $\boldsymbol{\beta_R = 0.1}$, $\boldsymbol{V_{BE}(sat) = 0.8\,V}$, $\boldsymbol{V_{BE}(active) = 0.7\,V}$, $\boldsymbol{V_{CE}(Sat) = 0.1V}$ and $\boldsymbol{V_{OH} = 3.4V}$

(a)	CO1	Briefly **explain** the advantages of using the totem-pole output stage.	[4]
(b)	CO2	Assume, $v_X = v_Y = 5V$ and no loads are connected. **Find** all the currents in **mA** and voltages in **V** labeled in the **Driver** circuit of **figure 2** (including $\boldsymbol{vo}$).	[10]
(c)	CO2	Assume, $v_X = v_Y = 0.1V$, **Determine** $\boldsymbol{I_2, I_3, \grave{I}_1}$ and **estimate** the maximum number of fanout you can connect to the driver circuit in this case.	[6]

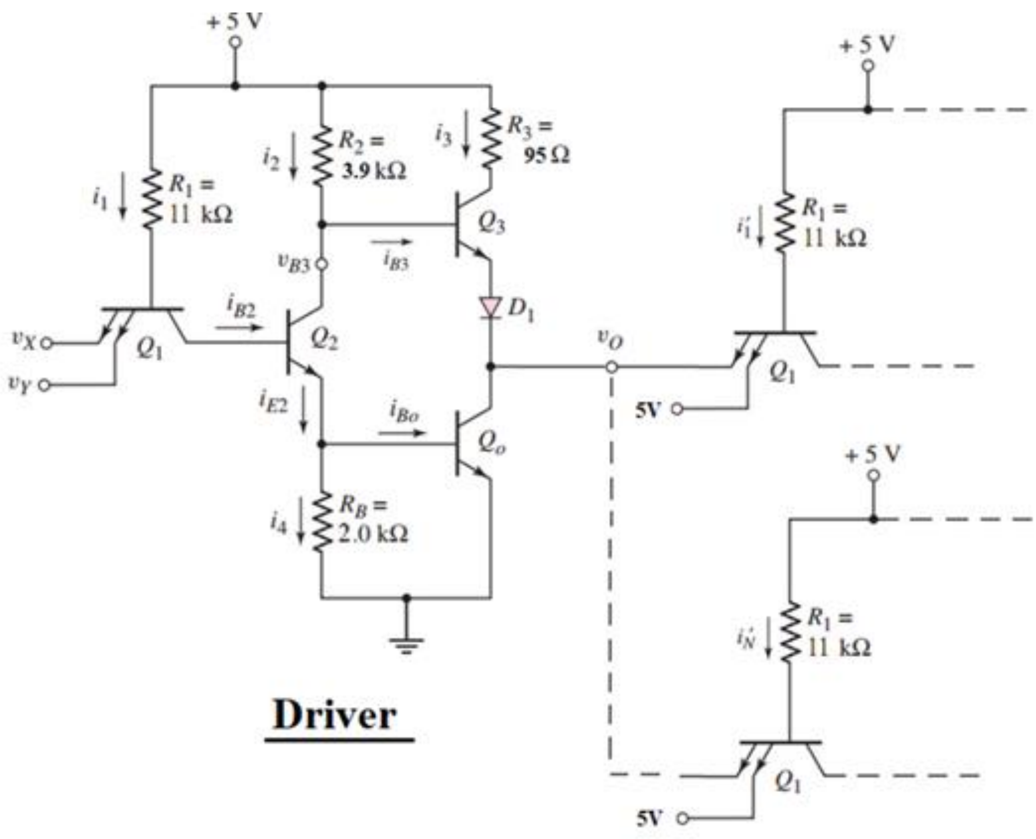

Figure 2

Question No. 3

For Figure 3 the transistor parameters are $K_n' = 100\ \mu A/V^2$ (use $K_n' * (W/L)$ to calculate K_n), $V_{TND} = V_{TNL} = 0.4\ V$, $(\frac{W}{L})_D = 16$, and $(\frac{W}{L})_L = 2$.

(a)	CO1	**Identify** the operating mode of the load NMOS transistor.	[4]
(b)	CO1	**Determine** the value of $\boldsymbol{V_O\ in\ V}$ **and** $\boldsymbol{I_{DL}, I_{DD}\ in\ mA}$ for $\boldsymbol{V_I = 0.1V\ and\ V_I = 1V}$	[8]
(c)	CO2	**Design** a static CMOS logic circuit that implements the logic function Y= (A+B)+CD .(indicate both **PMOS** and **NMOS** networks)	[4+4]

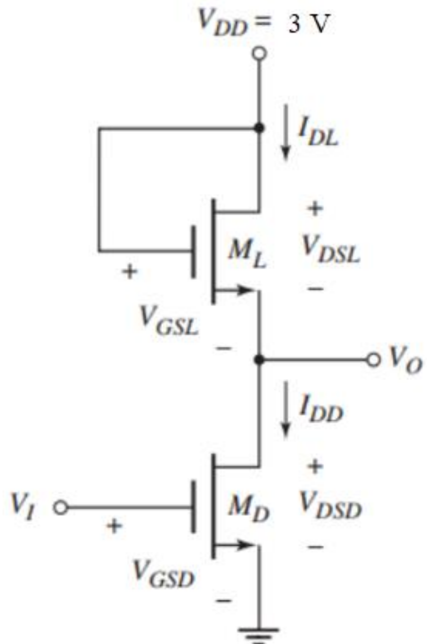

Figure 3

Question No. 4

Assume, V_{DD} = 2.5 V, K_n = 0.2 mA/V^2, K_p = 0.05 mA/V^2, V_{TN} = 0.6 V, V_{TP} = -1 V.

(a)	CO2	**Identify** the operating modes of PMOS and NMOS in **figure 4** at point A of the transfer characteristics curve. Then, **write** down the drain current **(I_D)** of PMOS and NMOS in terms of **V_I , V_O** and MOSFET parameters mentioned above.	[8]
(b)	CO2	**Calculate** the value of **V_{OHU}** and **V_{IL}** in volts.	[12]

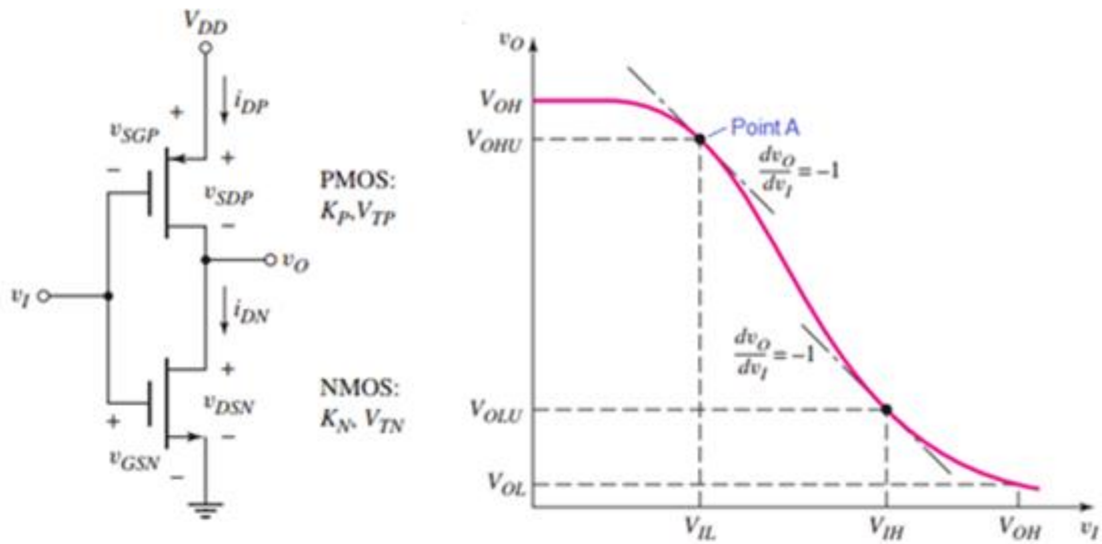

Figure 4

Question No. 5

In figure 5, **circuit no. 1** is known as the Inverting Schmitt Trigger, and the **circuit no. 2** is known as a square wave generator (astable multivibrator) that combines the principle of capacitive charging-discharging and the characteristics of a Schmitt trigger circuit.

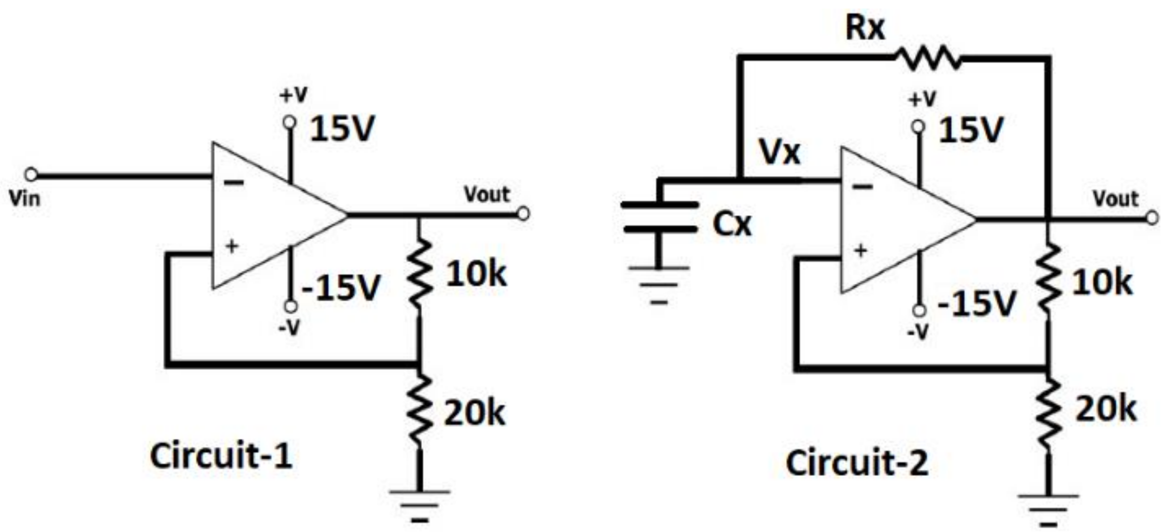

Figure 5

(a)	CO5	**Determine** the higher threshold voltage V_{TH} and lower threshold voltage V_{TL} of the given Schmitt Trigger. Finally **measure** the hysteresis width of this circuit.	[3+3+2]
(b)	CO5	**Given** $R_x = 100\ k\Omega\ and\ C_x = 0.621\ \mu F$, **find** the duration of the positive half cycle **T1**, and the negative half cycle **T2**. From the determined values of **T1** and **T2**, determine the **oscillation frequency f** and the **duty cycle D** of the astable multivibrator. **Comment** on how the oscillation frequency changes when you double the value of R_x.	[3+3+2+2+2]

Question No. 6

(a)	CO3	**Design** a 2-bit flash A/D converter circuit to convert an analog signal in the range of $0\ V\ to\ 5\ V$. **Calculate** total number of resistors and comparators required to fabricate this converter and **draw** the circuit. [**Hint:** the quantization levels are $0.625V, 1.875V, 3.125V\ and\ 4.375V$]	**[4+6]**
(b)	CO3	If at any time instance the input voltage is $\mathbf{3.54\ V}$, **find out** the 2-bit binary output obtained from the converter circuit designed in (a).	**[4]**
(c)	CO3	**Determine** the output voltage, v_O of the 4-bit weighted-resistor D/A in the **figure 6** for input = **0111**. [Hint: $v_O = V_R \sum \frac{b_n}{2^{n-1}}$]	**[6]**

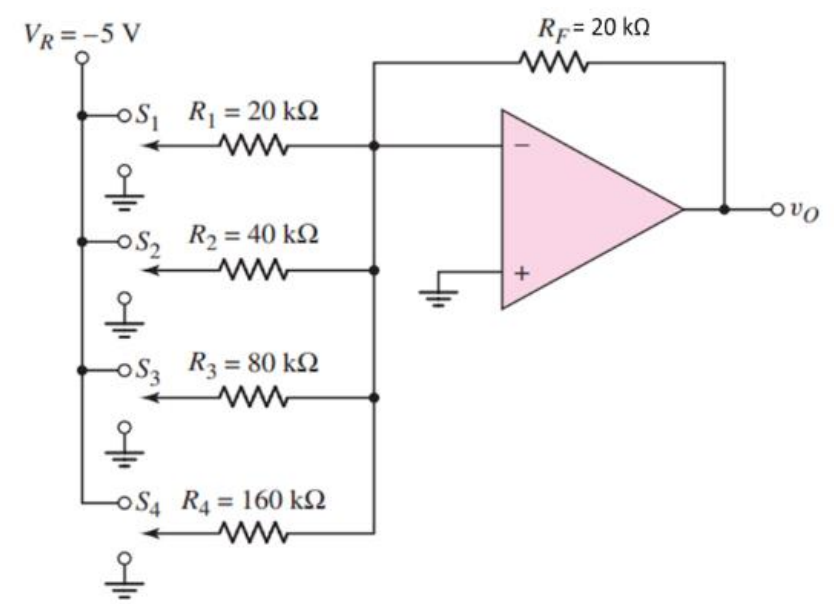

Figure 6